Detecting DDoS Attacks Through Decision Tree Analysis: An EDA Approach with the CIC DDoS 2019 Dataset

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***Abstract*— Distributed Denial of Service (DDoS) attacks pose a significant threat to network security, often causing severe service disruptions. This study explores the CIC DDoS 2019 dataset through Exploratory Data Analysis (EDA) to identify patterns and features critical for DDoS attack detection. The focus is on enhancing detection mechanisms using the Decision Tree algorithm, known for its simplicity, interpretability, and ability to manage both numerical and categorical data. The algorithm's transparency allows for clear insight into the decision-making process, making it easier to understand and explain classifications. Moreover, Decision Trees are robust against outliers, ensuring reliable performance in diverse network conditions. Key features and trends identified through EDA help differentiate between normal and malicious traffic. The Decision Tree algorithm effectively classifies and predicts DDoS attacks, achieving a ROC-AUC score of 99.13% and an accuracy of 98.55%. These findings underscore the algorithm's potential for real-time DDoS detection and mitigation. This research highlights the value of Decision Trees in cybersecurity, offering insights that can lead to more effective and adaptive network security strategies.**

Keywords—DDoS, Decision Tree, Machine Learning, Attack Detection, Traffic Malicious

I. Introduction

DDoS attacks represent a critical threat to network integrity and service availability [1]. By inundating a target system with an excessive volume of malicious traffic, these attacks aim to disrupt legitimate access and service operations [2]. The primary objective of a DDoS attack is to exhaust the resources of the target, such as bandwidth, processing power, or memory, rendering it incapable of responding to valid user requests [3]. This can lead to significant disruptions in service, affecting everything from online retail platforms to essential infrastructure services. The severity of such disruptions highlights the importance of understanding and addressing the impact of DDoS attacks on organizational operations and service continuity [4].

The challenge posed by DDoS attacks is further exacerbated by their increasing frequency and sophistication [5]. Modern DDoS attacks often employ advanced techniques, including distributed networks of compromised devices, to amplify their impact and evade detection. These attacks can vary in scale from smaller, targeted assaults to large-scale, multi-vector attacks that simultaneously exploit multiple vulnerabilities [6]. Organizations are consequently faced with the daunting task of defending against a constantly evolving threat landscape. The financial implications of successful DDoS attacks can be severe, encompassing direct costs such as downtime and indirect costs like loss of customer trust and market share [7].

Machine learning has revolutionized the approach to DDoS detection by providing advanced analytical capabilities that surpass those of traditional methods [8]. Machine learning algorithms can sift through vast amounts of network traffic data to identify subtle and complex patterns that might signal an attack [9]. These algorithms are designed to adapt and learn from historical data, allowing them to detect evolving threats and anomalies with greater accuracy [10]. By applying machine learning techniques, organizations can enhance their ability to detect and respond to DDoS attacks more effectively, reducing the likelihood of successful disruptions [11]. Among the various machine learning algorithms available, the Decision Tree algorithm has gained prominence for its simplicity, interpretability, and effectiveness in classification tasks [12].

The Decision Tree algorithm offers several advantages that make it particularly well-suited for DDoS attack detection [12]. Unlike more complex algorithms, Decision Trees create a clear and understandable model by recursively splitting data based on feature values, forming a tree-like structure of decisions [13]. Decision tree simplicity allows for easy interpretation of the decision-making process, making it easier to understand how certain features contribute to the classification of network traffic as normal or malicious. By utilizing historical attack data, Decision Trees build models that predict the likelihood of an attack, providing a robust method for enhancing network security [14]. This approach not only aids in identifying potential threats but also helps in developing proactive measures to mitigate the impact of DDoS attacks, thereby strengthening overall network resilience [15].

The intersection of machine learning technology and the problem of DDoS attacks represents a critical area of research with significant implications for cybersecurity [16]. As DDoS attacks become increasingly sophisticated and frequent, traditional methods for detection and mitigation often struggle to keep pace. Machine learning offers a transformative approach by enabling the analysis of large volumes of network traffic data in real-time [17]. This capability allows for the identification of abnormal patterns and behaviors that are indicative of DDoS attacks, which can be crucial for timely and effective response [18]. Machine learning algorithms can learn from historical data to recognize subtle and complex attack signatures, thereby improving the overall effectiveness of DDoS detection systems [19].

The CIC DDoS 2019 dataset plays a vital role in this research area, providing a comprehensive and detailed collection of network traffic data, including a wide range of DDoS attack types [20]. This dataset serves as a valuable resource for applying and evaluating machine learning techniques. It includes various attack scenarios and normal traffic patterns, offering a rich basis for training and testing machine learning models. By analyzing this dataset, researchers can gain deeper insights into the characteristics and dynamics of DDoS attacks, which are essential for developing effective detection and mitigation strategies. The breadth and depth of the data facilitate a thorough examination of how different features and attack types impact model performance [21].

The primary objective of this study is to advance the detection of DDoS attacks by applying the Decision Tree algorithm to the CIC DDoS 2019 dataset. This research focuses on leveraging Decision Trees, a machine learning technique known for its clarity and effectiveness, to improve the identification and classification of DDoS attacks. By performing a comprehensive exploratory analysis of the dataset, the study aims to uncover key features and patterns that are indicative of DDoS activity [19]. This involves analyzing the dataset to understand which network traffic characteristics are most predictive of attacks, thus enhancing the ability of the Decision Tree algorithm to distinguish between legitimate and malicious traffic.

This study represents a significant advancement in the field of network security by introducing a novel approach to detecting DDoS attacks through the application of Decision Tree analysis. While prior research has explored various machine learning techniques for DDoS detection, this study uniquely leverages the simplicity, transparency, and robustness of the Decision Tree algorithm in the context of EDA using the CIC DDoS 2019 dataset. The novelty lies in the methodical integration of EDA to identify critical features and trends within network traffic data, which are then utilized to enhance the accuracy of DDoS detection. The study further seeks to evaluate the performance of the Decision Tree algorithm in real-world scenarios, assessing its efficacy in classifying and predicting DDoS attacks accurately. Through rigorous testing and validation, the research aims to demonstrate how well Decision Trees can operate in real-time environments, providing timely and reliable detection of DDoS threats. The insights gained from this study are expected to contribute significantly to the development of more robust and adaptive network security strategies, supporting the creation of resilient digital infrastructures better equipped to defend against the growing and evolving threat of DDoS attacks.

II. Related Works

The problem of DDoS attacks has garnered significant attention in recent research due to its escalating impact on network security [22]. Early studies by [23] and [24] focused on traditional mitigation techniques, such as rate limiting and firewalls, which often proved inadequate against the increasing scale and complexity of DDoS attacks. Research has documented the evolution of attack methods, including the use of botnets and amplification techniques, which have amplified the challenges faced by organizations. Studies like those by [25] illustrate how the frequency and severity of DDoS attacks have grown, leading to substantial financial and operational impacts. More recent works have highlighted the need for advanced detection and mitigation strategies as traditional methods struggle to cope with sophisticated and large-scale attacks.

As machine learning technologies have advanced, they have been increasingly applied to the problem of DDoS detection and mitigation [26]. Research has shown that machine learning algorithms can significantly improve the identification of abnormal traffic patterns associated with DDoS attacks. Studies such as those by [27] and [28] explore various machine learning techniques, including clustering and classification algorithms, to enhance DDoS detection capabilities.

The Decision Tree algorithm has been widely studied for its effectiveness in various classification tasks, including network security [29]. Research by [30] established the foundational principles of Decision Trees, demonstrating their simplicity and interpretability. Recent studies by [31] have explored the application of Decision Trees specifically in the context of DDoS attack detection. For example, works by [27] highlight how Decision Trees can be utilized to analyze network traffic and classify attack patterns effectively. These studies show that Decision Trees can provide clear decision rules and offer insights into the features most indicative of DDoS attacks, enhancing the accuracy and reliability of detection systems.

Several recent studies have investigated similar approaches to using Decision Trees for DDoS attack detection [32]. For instance, research by [33] has applied Decision Tree algorithms to large-scale network datasets, demonstrating their effectiveness in distinguishing between normal and malicious traffic. These studies highlight the algorithm's ability to handle complex datasets and provide actionable insights into attack detection. The findings suggest that Decision Trees, when combined with comprehensive datasets like CIC DDoS 2019, can offer robust solutions for real-time DDoS detection and contribute to the development of more resilient cybersecurity strategies.

In reviewing existing literature on DDoS detection, multiple researchers have explored various algorithms with differing results. Research by [34] on the MultiTree algorithm achieved an accuracy of 84.23%, illustrating its moderate success in classification tasks. The K-Nearest Neighbors (KNN) algorithm, studied by [15][35], showed a higher accuracy of 89.0%, indicating its effectiveness in pattern recognition. Furthermore, [36] employed Convolutional Neural Networks (CNN) and reported an accuracy of 87.0%, highlighting CNN's potential in handling complex data. Meanwhile, [37] explored the Decision Table algorithm, achieving an accuracy of 88.43%, showcasing its utility in decision-making processes. These varied outcomes underscore the ongoing quest for optimizing DDoS detection methods, with each algorithm offering unique strengths and limitations in cybersecurity contexts.

III. Material and Methods

This study leverages the CIC DDoS 2019 dataset to enhance the detection of DDoS attacks using the Decision Tree algorithm. The following steps outline the materials and methods used in this research show in Figure 1.

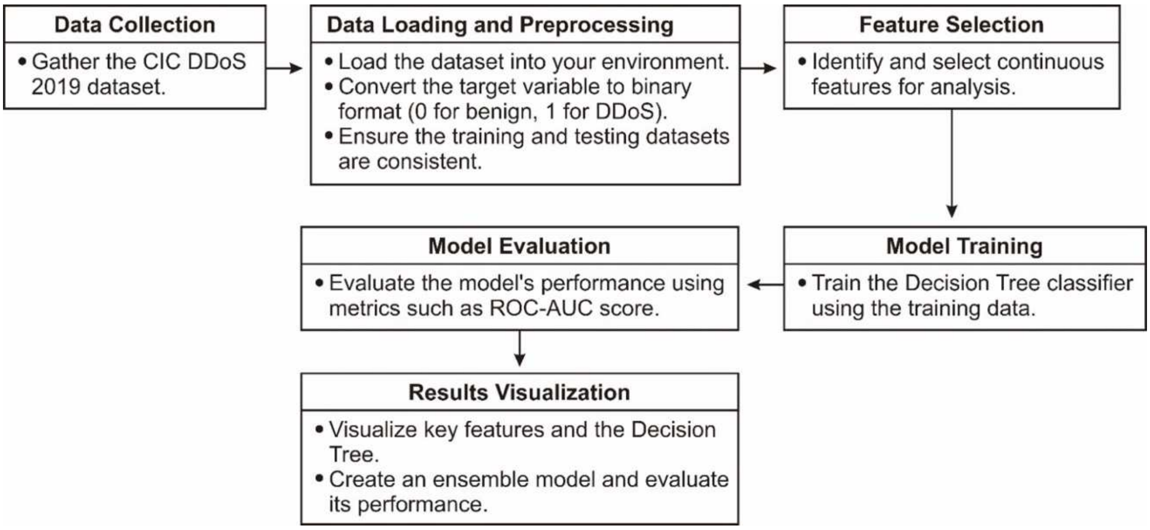

Figure 1. The block diagram of the proposed method

A. Data Collection and Preparation

The dataset used in this study is the CIC DDoS 2019 dataset from [20], which contains various types of DDoS attack traffic and benign traffic. The dataset is stored in multiple Parquet files, separated into training and testing sets. The script begins by importing essential libraries such as numpy, pandas, and os.

To load the data, the script traverses the directory structure to collect file paths for training and testing datasets. The os.walk function is used to iterate through the directory and gather the file paths based on their suffixes '-training' and '-testing'.

B. Data Loading and Preprocessing

The collected Parquet files are loaded into pandas DataFrames and concatenated to form the complete training and testing datasets. This ensures that all relevant data is combined into a single structure for each set. The dataset used in this study consists of both training and testing data, which have been concatenated from multiple parquet files. The training data, stored in df_train, comprises 125,170 samples, each described by 78 features. Similarly, the testing data, stored in df_test, contains 306,201 samples, also described by 78 features.

The target variable, 'Label', is converted to a binary format where 'Benign' traffic is labeled as 0 and any DDoS attack is labeled as 1. This binary classification simplifies the task of identifying attack traffic versus normal traffic. The training and testing datasets exhibit distinct distributions of labels, indicating the variety of DDoS attacks and benign traffic they contain. In the training dataset, labeled \`df_train\`, the most prevalent categories are 'Syn' attacks with 48,840 samples, followed closely by 'Benign' traffic with 46,427 samples. Other significant attack types include 'UDP' with 18,090 samples and 'MSSQL' with 8,523 samples. Lesser-represented categories include 'LDAP' with 1,906 samples, 'Portmap' with 685 samples, 'NetBIOS' with 644 samples, and 'UDPLag' with 55 samples. This distribution showcases a diverse array of attack types that the model will be trained on show in Table 1.

In contrast, the testing dataset, labeled \`df_test\`, contains a different distribution of labels, reflecting a wider variety of DDoS attack types. The most significant categories are 'DrDoS_NTP' with 121,368 samples and 'TFTP' with 98,917 samples. 'Benign' traffic is the third most prevalent with 51,404 samples. Other notable categories include 'DrDoS_UDP' with 10,420 samples, 'UDP-lag' with 8,872 samples, and 'DrDoS_MSSQL' with 6,212 samples. Additionally, there are smaller quantities of 'DrDoS_DNS' with 3,669 samples, 'DrDoS_SNMP' with 2,717 samples, 'DrDoS_LDAP' with 1,440 samples, 'DrDoS_NetBIOS' with 598 samples, 'Syn' with 533 samples, and 'WebDDoS' with 51 samples show in Table 1.

Table 1. Training and testing datasets

Train Label		Test Label	
Syn	48840	DrDoS_NTP	121368
Benign	46427	TFTP	98917
UDP	18090	Benign	51404
MSSQL	8523	DrDoS_UDP	10420
LDAP	1906	UDP-lag	8872
Portmap	685	DrDoS_MSSQL	6212
NetBIOS	644	DrDoS_DNS	3669
UDPLag	55	DrDoS_SNMP	2717
		DrDoS_LDAP	1440
		DrDoS_NetBIOS	598
		Syn	533
		WebDDoS	51
Name : count, dtype : int64		Name : count, dtype : int64	

C. *Feature Selection and Evaluation*

Continuous features are identified, excluding the target variable. The script includes a helper function, xs_y, to separate the features (X) from the target variable (y). This function ensures that the features and target are correctly partitioned for model training and evaluation. An initial evaluation examines the distribution of 'Bwd Packets/s' by grouping the data based on the target variable, providing insights into feature behavior relative to attack labels.

D. *Model Training and Evaluation*

The Decision Tree classifier is trained using a custom function, evaluate_one_feature, which evaluates the performance of the classifier on individual features using the ROC-AUC score. This function trains a Decision Tree on each feature and assesses its ability to distinguish between attack and benign traffic. The decision tree generated using the DecisionTreeClassifier from scikit-learn, specifically focused on the "Bwd Packets/s" feature, illustrates how the dataset is split to classify samples as either benign (0) or DDoS attacks (1).

Figure 2 show root node of the tree splits the data based on the condition `x[0] <= 0.236`, where `x[0]` represents the "Bwd Packets/s" feature. At this node, the Gini impurity is 0.467, with a total of 125,170 samples, comprising 46,427 benign samples and 78,743 attack samples. The left child node, resulting from the split, has a lower Gini impurity of 0.219 and contains 80,309 samples, with 10,067 benign and 70,242 attack samples. Conversely, the right child node has a Gini impurity of 0.307, with 44,861 samples, including 36,360 benign and 8,501 attack samples.

This structure demonstrates that the threshold value of the "Bwd Packets/s" feature effectively differentiates between benign and attack samples, with the Gini index values indicating the purity of the splits. Overall, this decision tree highlights the significance of the "Bwd Packets/s" feature in identifying DDoS attacks, showing how well the splits segregate the samples into benign and attack categories.

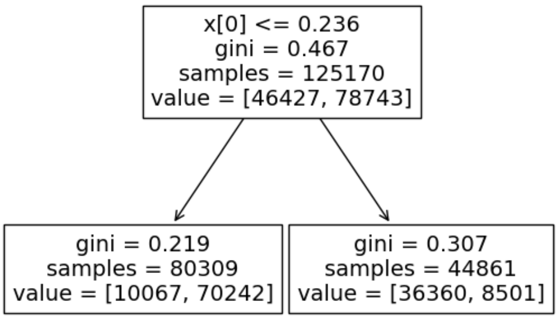

Figure 2. Decision Tree Based on 'Bwd Packets/s'

IV. THE EXPERIMENTAL RESULTS

A. *Visualization*

Top features identified through ROC-AUC scores are visualized using seaborn and matplotlib. An ensemble model is created by averaging the predictions of these top-performing features. The optimal threshold for classification is determined using the Youden’s J statistic. The final model's performance is evaluated using various metrics, including ROC-AUC, precision, recall, F1-score, and accuracy. The classification results are visualized using a confusion matrix.

B. *Final Model Evaluation*

In summary, this detailed approach combines data preprocessing, feature selection, model training, and evaluation to effectively utilize the Decision Tree algorithm for detecting DDoS attacks in the CIC DDoS 2019 dataset.

The evaluation metrics for the Decision Tree model used for DDoS attack detection are impressive. The confusion matrix indicates a high level of accuracy, with most predictions falling on the diagonal, signifying correct classifications. Specifically, the model correctly classified 48,993 instances as negatives and 252,789 instances as positives, with only a small number of misclassifications (2,411 false positives and 2,008 false negatives) show in figure 3.

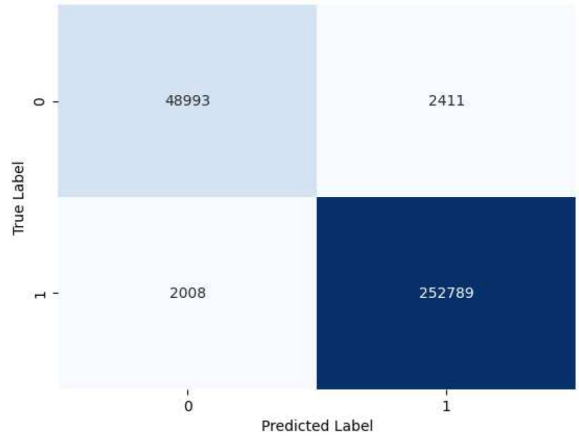

Figure 3. Confusion Matrix

Table 2 presents a comprehensive set of performance metrics evaluating the Decision Tree model's effectiveness in detecting DDoS attacks. The precision of 99.06% indicates that when the model predicts an instance as a DDoS attack, it is correct 99.06% of the time, highlighting its ability to minimize false positives. The recall, at 99.21%, measures the model's capability to identify 99.21% of all actual DDoS attacks, ensuring that most DDoS attacks are detected and not missed. The F1-score, which is the harmonic mean of precision and recall, stands at 99.13%, reflecting a balanced and high performance in both identifying DDoS attacks and ensuring accurate positive predictions. The model's overall accuracy is 98.55%, indicating that 98.55% of all predictions (both positive and negative) were correct. Additionally, the ROC-AUC score of 99.13% signifies the model's excellent ability to distinguish between DDoS attacks and benign traffic across all classification thresholds. These metrics collectively demonstrate that the Decision Tree model performs exceptionally well in detecting DDoS attacks, with high precision, recall, accuracy, and robust discrimination capability.

Table 2. Result Evaluation

Precision	99,06%
Recall	99,21%
F1-Score	99,13%

Accuracy	98,55%
ROC-AUC	99,13%

C. Comparative Research

The table 3 summarizes the performance of various algorithms in detecting DDoS attacks, as researched by different scholars. MultiTree, investigated by [34], achieved an accuracy of 84.23%. KNN, examined by [35], attained 89.0%, while CNN, analyzed by [36], reached 87.0%. Decision table, studied by [37], resulted in an accuracy of 88.43%. In comparison, our study utilizing the Decision Tree algorithm significantly outperformed these methods, achieving an impressive accuracy of 98.55%. This highlights the superior effectiveness of the Decision Tree algorithm in this context.

Table 3. Comparative Research

Researcher	Algorithm	Accuracy
[34]	MultiTree	84.23%
[35]	KKN	89.0%
[36]	CNN	87.0%
[37]	Decision Table	88.43%
Our Study	**Decision Tree**	**98.55%**

V. CONCLUSION

Based on the comprehensive evaluation of the Decision Tree model for DDoS attack detection, several significant conclusions can be drawn. Firstly, the model demonstrates exceptional performance, as evidenced by the high precision of 99.06%, indicating its effectiveness in correctly identifying DDoS attacks with minimal false positives. The recall rate of 99.21% further highlights the model's capability to detect the vast majority of actual DDoS attacks, ensuring that very few incidents go unnoticed. The balanced performance is reflected in the F1-score of 99.13%, which combines both precision and recall into a single metric. Additionally, the model's overall accuracy of 98.55% signifies that it reliably distinguishes between benign and malicious traffic in most instances. The high ROC-AUC score of 99.13% underscores the model's robust discriminatory power across all classification thresholds. Collectively, these results indicate that the Decision Tree algorithm is highly effective for real-time DDoS attack detection and mitigation. This study thus supports the use of Decision Tree models as a reliable tool in enhancing network security, providing a foundation for the development of more resilient and adaptive strategies against DDoS threats.

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